PROTOCOL FOR PHYSICAL AND DATA-LINK LAYER ENCODING NEGOTIATION

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Title: PROTOCOL FOR PHYSICAL AND DATA-LINK LAYER ENCODING NEGOTIATION

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Abstract

A communication protocol for negotiating data encoding schemes at the Physical (Layer 1) and Data-Link (Layer 2) layers of the OSI model, specifically for high-cardinality parallel interfaces. The protocol comprises: a device, upon connection, transmitting a capabilities frame using a universally supported, low-cardinality default encoding; the capabilities frame advertising the device's supported encoding maps (e.g., "UTF-8-Retrofit," "Machine-Code-24bit"); a receiving device responding with a selection frame choosing one of the advertised encodings; and both devices then switching from the default encoding to the selected, high-density encoding for all subsequent data transmission in the session. This moves encoding negotiation from the Application Layer to the hardware layer, a necessary innovation for interfaces where the physical signal representation itself is variable.

Background of the Invention

In conventional networking, the physical layer is fixed and binary; it transmits 1s and 0s. The interpretation of these bits into characters or data structures is handled at higher levels of the OSI model, most notably the Application Layer (Layer 7). For example, the HTTP protocol uses headers to declare the character encoding of the content (e.g., 'Content-Type: text/html; charset=UTF-8'). This approach is sufficient when the physical layer is dumb, but it is inadequate for new high-cardinality physical layers, where the "language" of the physical signals can change. There is no existing protocol to allow hardware to negotiate the meaning of its own physical states before data transmission begins.

Summary of the Invention

The present invention is a protocol, or handshake process, that allows two connected devices to dynamically negotiate and agree upon a data encoding map at the Physical and Data-Link layers (Layers 1 and 2). This enables the use of high-cardinality interfaces where the meaning of a physical state is not fixed.

The protocol operates as follows:

1. **Initialization:** When two devices are physically connected, they begin communication using a pre-defined, "default" encoding map. This default map is a low-density, universally supported standard (e.g., mapping a single bit to a black or white pixel state) to guarantee the initial handshake can be understood.

2. **Capabilities Advertisement:** The initiating device sends a "capabilities frame" across the link using this default encoding. This frame contains a list of all the high-density encoding maps it supports (e.g., "Standard Machine Code 24-bit," "UTF-8 Retrofit," "Session-Specific Disposable Map XYZ").

3. **Selection and Acknowledgment:** The receiving device reads the capabilities frame and selects a compatible encoding map from the list that it also supports. It sends a "selection frame" back to the initiating device, indicating its choice.

4. **Protocol Switch:** Once the selection is acknowledged, both devices load the selected high-density encoding map and switch to using it for the remainder of the communication

session. All subsequent frames are interpreted according to this negotiated map.

This lower-layer negotiation is a fundamental requirement for unlocking the potential of variable-encoding hardware.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a sequence diagram illustrating the steps of the lower-layer encoding negotiation protocol.

Detailed Description of the Invention

Referring to **FIG. 1**, a sequence diagram shows the negotiation process between two devices, Device A (101) and Device B (102), connected by a high-cardinality parallel link. First, a physical connection is established (103). Immediately following, Device A sends a capabilities frame (104) to Device B. This frame is encoded using a universal default, low-density map and lists the high-density maps supported by Device A. Device B receives this frame and compares the list to its own supported maps. Device B then sends a selection frame (105) back to Device A, specifying the chosen encoding map for the session. Upon receiving the selection, Device A sends an acknowledgment (ACK) frame (106). Once the handshake is complete, both devices switch their physical layer encoders and decoders to use the selected map, and the main data transmission begins (107) using the efficient, high-density encoding.

Claims

What is claimed is:

1. A method for negotiating a data encoding scheme between two devices connected by a parallel communication interface, the method comprising the steps of:

    - establishing a communication session using a default, pre-defined encoding map;
    - transmitting, from a first device to a second device, a capabilities message listing one or more available high-density encoding maps;
    - transmitting, from the second device to the first device, a selection message indicating a chosen encoding map from the list; and
    - conducting subsequent data transmission for the session using the chosen high-density encoding map.

2. The method of claim 1, wherein the negotiation occurs at the Physical Layer (Layer 1) or Data-Link Layer (Layer 2) of the OSI model.

3. The method of claim 1, wherein the default encoding map is a low-cardinality map and the chosen encoding map is a high-cardinality map.

4. The method of claim 1, wherein the parallel communication interface comprises a two-dimensional array of pixels.

5. The method of claim 4, wherein the default encoding map represents a binary state using black and white pixel values.

6. The method of claim 4, wherein the chosen high-density encoding map maps multi-byte data blocks to unique pixel color values.

7. The method of claim 1, wherein the parallel communication interface is a massively parallel optical interface as described in U.S. Patent Application No. 19/559,705.

Drawings

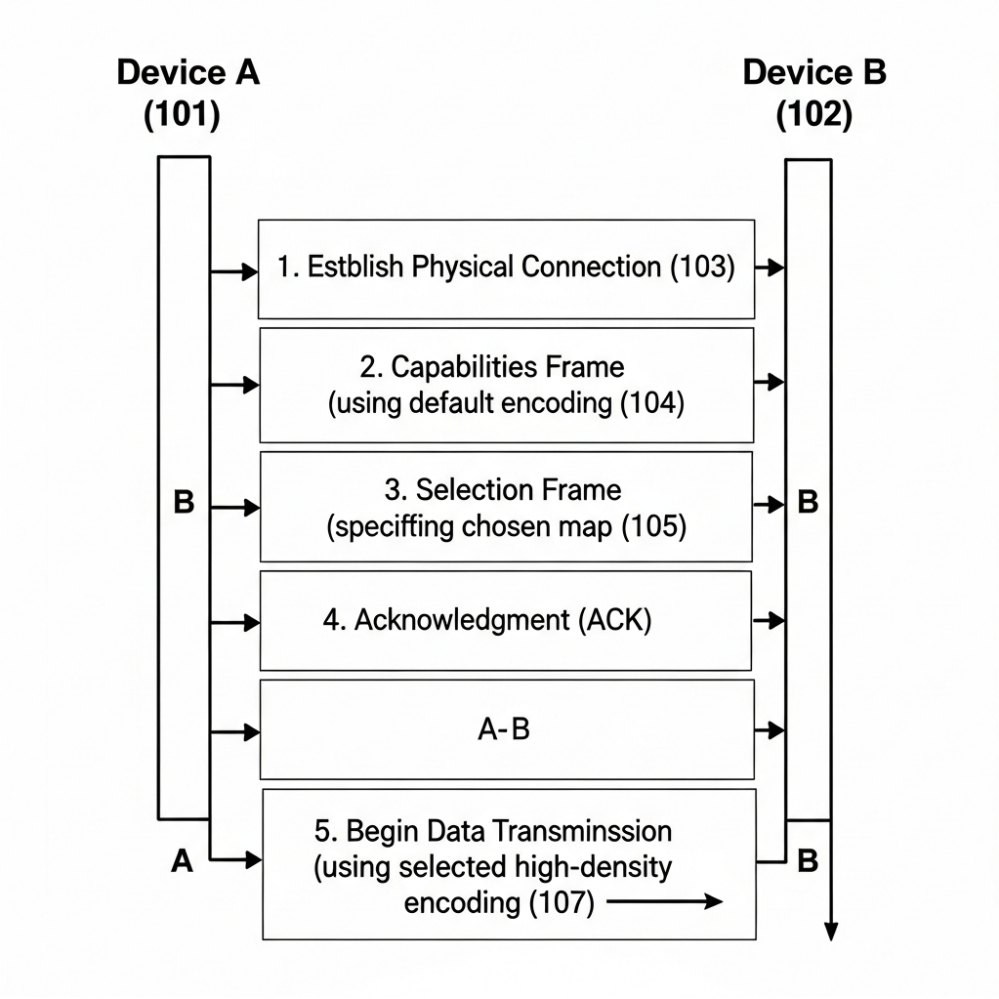


Figure 1: Sequence diagram illustrating the steps of the lower-layer encoding negotiation protocol.